HARD · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

One-variable data: Distributions and measures of center and spread

02

10 answers · Rationale-rich · Teach from doc

Q1 B

B. II only

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. The histogram shows that all the values in data set A are greater than 18. Since the additional number of points, 18, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with 50 values is between the 25th and 26th values when the values are listed in ascending order. The median of a data set with 51 values is the 26th value when the values are listed in ascending order. Therefore, the median of data set B is the 26th value in data set B, which is the 25th value in data set A. The histogram shows that the 25th and 26th values in data set A are both between 50 and 60 points. Therefore, the 25th and 26th values in data set A could be equal, in which case the median of data set A would be equal to both the 25th and 26th values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

4

The correct answer is 4. It's given that the two boxplots show the distribution of number of books read over the summer by the students in two different English classes. In a boxplot, the first vertical line represents the minimum value of the data set and the last vertical line represents the maximum value of the data set. The range of a data set is the difference between its maximum value and its minimum value. In class A, the maximum number of books read is 5 and the minimum number of books read is 0. The difference between those values is 5 - 0, or 5. Therefore, the range of the number of books read in class A is 5. In class B, the maximum number of books read is 10 and the minimum number of books read is 1. The difference between those values is 10 - 1, or 9. Therefore, the range of the number of books read in class B is 9. To find the positive difference between the ranges of the number of books read for the two classes, the smaller range must be subtracted from the larger range. Therefore, the positive difference between the ranges of number of books read over the summer for the two classes is 9 - 5, or 4.

Q3

1

The correct answer is 1. When there are an odd number of values in a data set, the median of the data set is the middle number when the data values are ordered from least to greatest. The scores for group A, ordered from least to greatest, are 2, 3, 4, 4, 5, 6, 6, 6, and 9. The median of the scores for group A is therefore 5. The scores for group B, ordered from least to greatest, are 5, 5, 5, 5, 6, 6, 8, 8, 9, 10, and 10. The median of the scores for group B is therefore 6. The median score for group B is $6-5=1$ more than the median score for group A.

Q4

41

The correct answer is 41. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of 75 objects and has a mean of 25 meters. This can be represented by the equation $\frac{x}{75} = 25$, where x represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by 75 yields $x = 7525$, or $x = 1{,}875$ meters. Therefore, the sum of the heights of the objects in data set A is $1{,}875$ meters. It's also given that data set B consists of the heights of 50 objects and has a mean of 65 meters. This can be represented by the equation $\frac{y}{50} = 65$, where y represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by 50 yields $y = 5065$, or $y = 3{,}250$ meters. Therefore, the sum of the heights of the objects in data set B is $3{,}250$ meters. Since it's given that data set C consists of the heights of the 125 objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1{,}875 + 3{,}250}{125}$, which is equivalent to 41 meters. Therefore, the mean, in meters, of data set C is 41.

Q5

B

B. 0.95%

Choice B is correct. The median of a set of numbers is the middle value of the set values when ordered from least to greatest. If the percents in the table are ordered from least to greatest, the middle value is 27.9%. The difference between 27.9% and 26.95% is 0.95%.

Choice A is incorrect and may be the result of calculation errors or not finding the median of the data in the table correctly. Choice C is incorrect and may be the result of finding the mean instead of the median. Choice D is incorrect and may be the result of using the middle value of the unordered list.

Q6

A

A.

Value	Frequency
70	4
80	5
90	6
100	7

Choice A is correct. The tables in choices B, C, and D each represent a data set where the values 80 and 90 have the same frequency and the values 70 and 100 have the same frequency. It follows that each of these data sets is symmetric around the value halfway between 80 and 90, or 85. When a data set is symmetric around a value, that value is the mean of the data set. Therefore, the data sets represented by the tables in choices B, C, and D each have a mean of 85. The table in choice A represents a data set where the value 90 has a greater frequency than the value 80 and the value 100 has a greater frequency than the value 70. It follows that this data set has a mean greater than 85. Therefore, of the given choices, choice A represents the data set with the greatest mean.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

A

A. The mean of data set A is greater than the mean of data set B.

Choice A is correct. The mean value of a data set is the sum of the values of the data set divided by the number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set A, the sum of products of each value and its frequency is $302 + 344 + 385 + 427 + 469$, or 1,094. It's given that there are 27 values in data set A. Therefore, the mean of data set A is $\frac{1,094}{27}$, or approximately 40.52. Similarly, the mean of data B is $\frac{958}{27}$, or approximately 35.48. Therefore, the mean of data set A is greater than the mean of data set B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

B

B. 90

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It's given that the mean length of the 240 gray seals from Muskeget Island was 88 inches. This can be represented by the equation $\frac{x}{240} = 88$, where x represents the sum of the lengths, in inches, of the 240 gray seals from Muskeget Island. Multiplying both sides of this equation by 240 yields $x = (240)(88)$, or $x = 21{,}120$ inches. It's also given that the mean length of the 120 gray seals from Sable Island was 94 inches. This can be represented by the equation $\frac{y}{120} = 94$, where y represents the sum of the lengths, in inches, of the 120 gray seals from Sable Island. Multiplying both sides of this equation by 120 yields $y = (120)(94)$, or $y = 11{,}280$ inches. Therefore, the sum of the lengths, in inches, of all 360 gray seals is $21{,}120 + 11{,}280$, or 32,400. Dividing this sum by 360 yields $\frac{32{,}400}{360}$, or 90 inches. Therefore, the mean length of all 360 gray seals the scientist measured for this study is 90 inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of 88 inches and 94 inches, not the mean length of all 360 gray seals the scientist measured for this study.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

C

C. The median mass of group 1 is greater than the median mass of group 2.

Choice C is correct. The median of a data set represented in a box plot is represented by the vertical line within the box. It follows that the median mass of the gazelles in group 1 is 25 kilograms, and the median mass of the gazelles in group 2 is 24 kilograms. Since 25 kilograms is greater than 24 kilograms, the median mass of group 1 is greater than the median mass of group 2.

Choice A is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice B is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 A

A. I only

Choice A is correct. The median of a data set with an odd number of values that are in ascending or descending order is the middle value of the data set. Since the distribution of the values of both data set A and data set B form symmetric dot plots, and each data set has an odd number of values, it follows that the median is given by the middle value in each of the dot plots. Thus, the median of data set A is 13, and the median of data set B is 13. Therefore, statement I is true. Data set A and data set B have the same frequency for each of the values 11, 12, 14, and 15. Data set A has a frequency of 1 for values 10 and 16, whereas data set B has a frequency of 2 for values 10 and 16. Standard deviation is a measure of the spread of a data set; it is larger when there are more values further from the mean, and smaller when there are more values closer to the mean. Since both distributions are symmetric with an odd number of values, the mean of each data set is equal to its median. Thus, each data set has a mean of 13. Since more of the values in data set A are closer to 13 than data set B, it follows that data set A has a smaller standard deviation than data set B. Thus, statement II is false. Therefore, only statement I must be true.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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